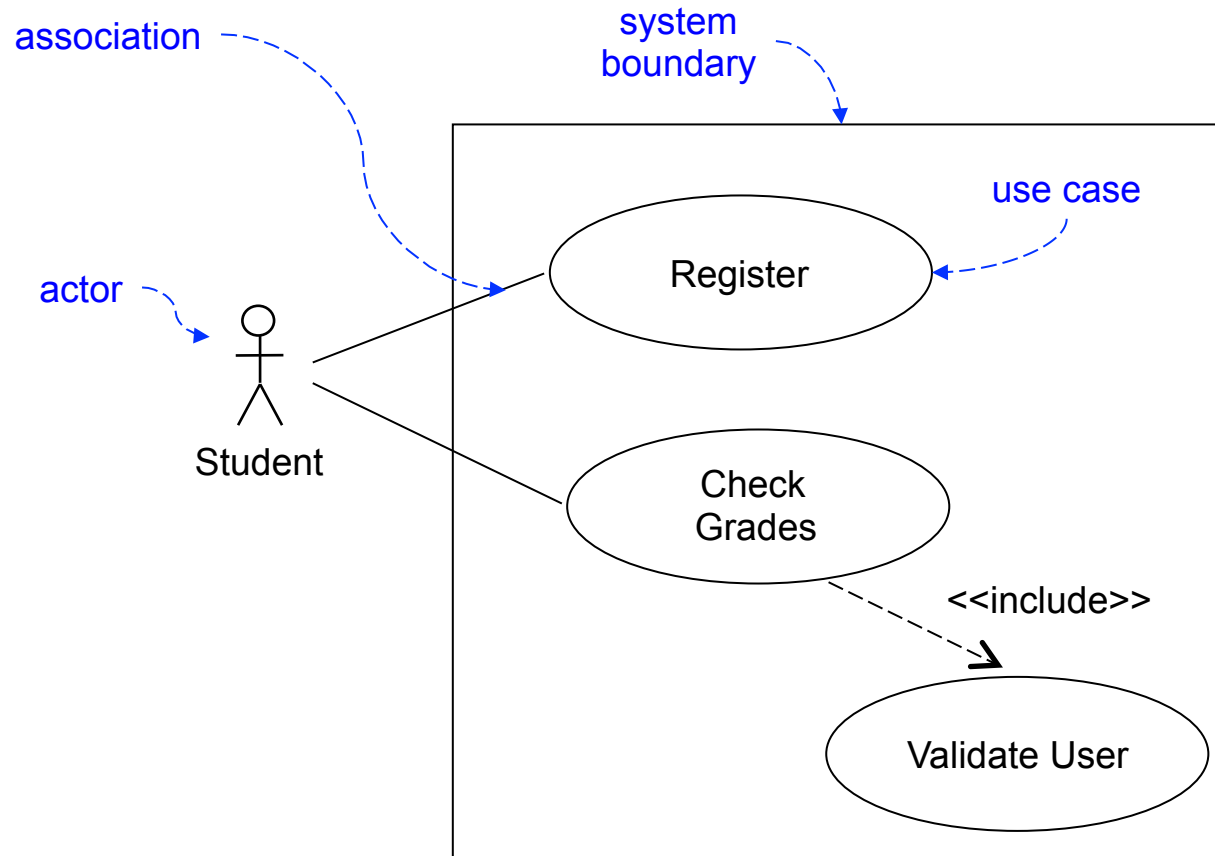


Use Case: elements



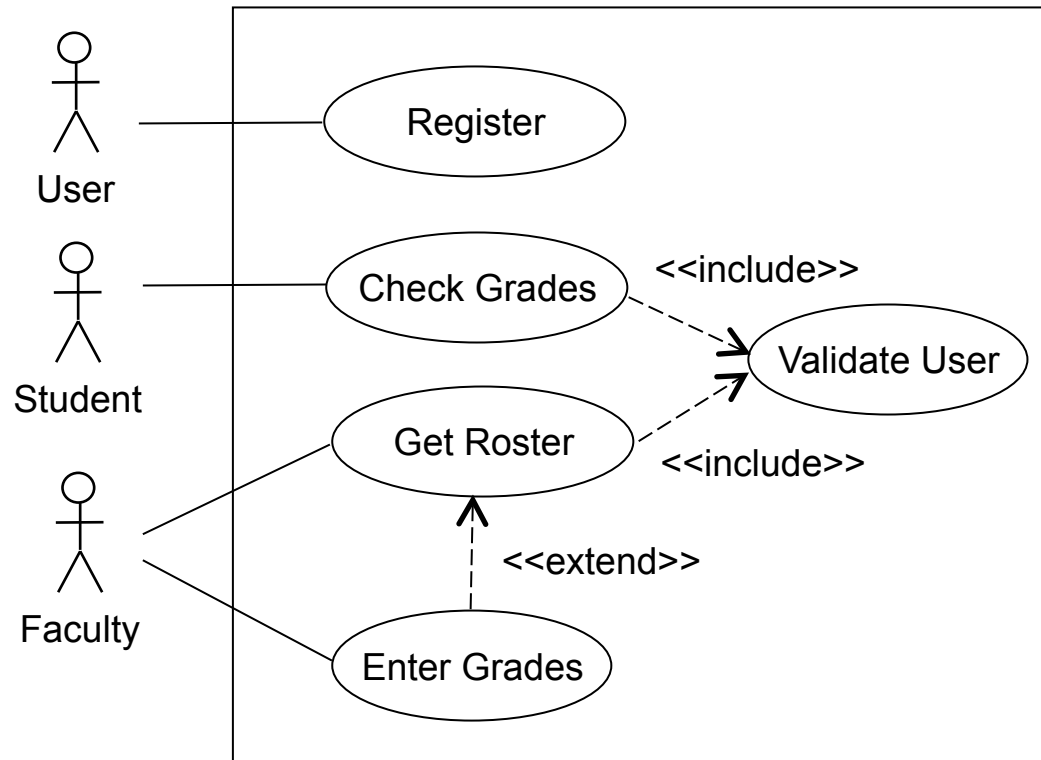
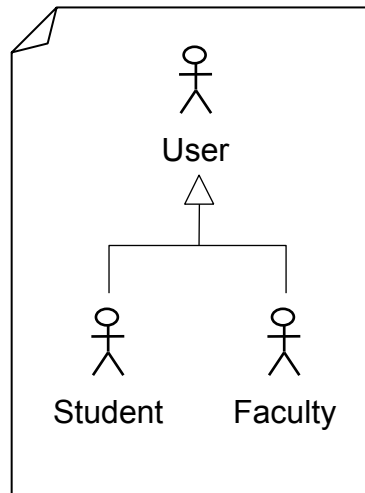
Elements of a Use Case Diagram

- **Actor:**
 - Represents a role played by external entities (humans, systems) that interact with the system
- **Use case:**
 - Describes what the system does (i.e., functionality)
 - Scenario: sequence of interactions between the actors and the system
- **Relationships:**
 - Association between actors and use cases
 - Extension (or generalization) among actors
 - Dependency among use cases: *include* and *extend*

Scenario / System Boundary

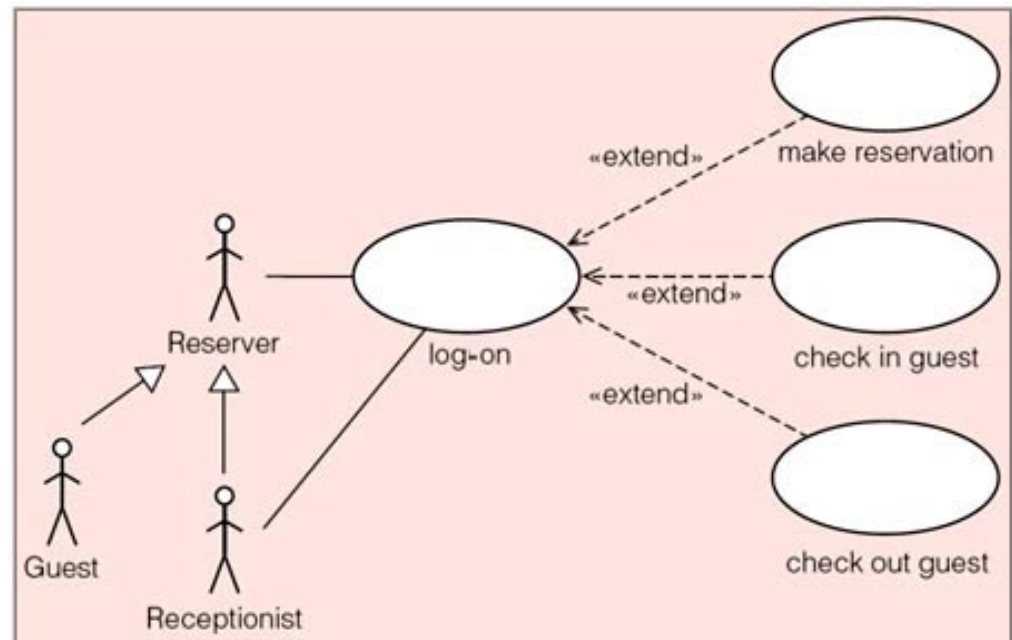
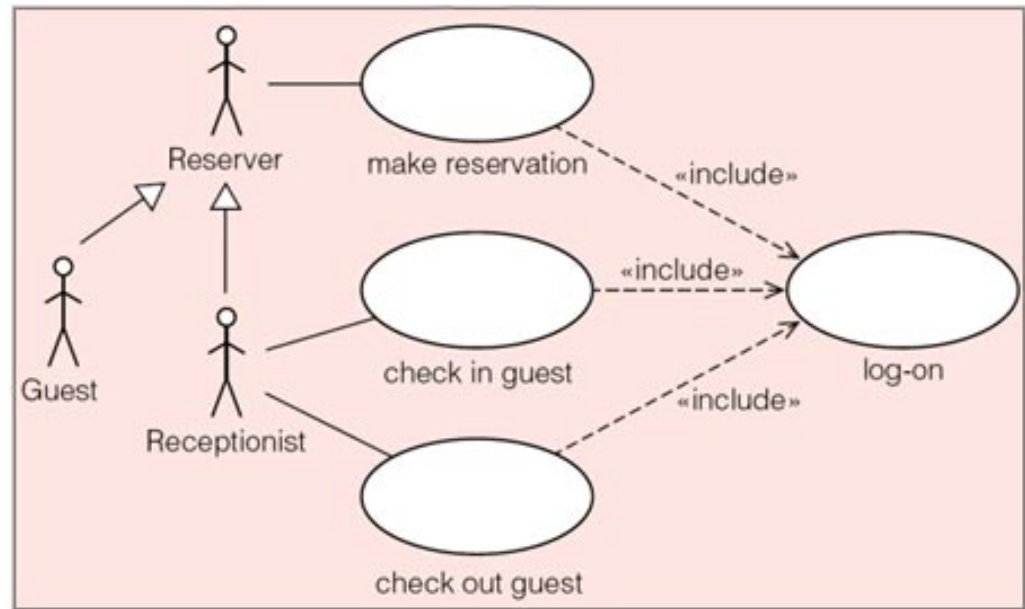
- **Scenario**
 - A single path through a Use Case
 - Use case is usually a collection of scenarios
 - Included as part of use case description
- **System Boundary**
 - A high level indication of the domain
 - Limit to investigation
 - System
 - Part of system in focus

Example



<<extend>> vs <<include>>

- A use case B is included in use cases C and D when these have some common steps represented by B
- A use case B extends a use case C when B applies optionally, under some condition (usually specified in the scenario)
- Note: the lower diagram is formally correct but should be avoided, because the main functions should NOT be described as extensions of logon



Use Case Scenario

Use Case: Check Grades	
Description: View the grades of a specific year and semester Actors: Student Precondition: The student is already registered Main scenario:	
User	System
3. The user enters the year and semester, e.g., Fall 2013.	1. The system carries out “Validate User”, e.g., for user “miner” with password “allAs”. 2. The system prompts for the year and semester. 4. The system displays the grades of the courses taken in the given semester, i.e., Fall 2013.
Alternative: The student enters “All” for the year and semester, and the system displays grades of all courses taken so far. Exceptional: The “Validate User” use case fails; the system repeats the validation use case.	